Reg. No. | | | | | | | | | | | [23MSCH1H02]

FIRST SEMESTER UNIVERSITY EXAMINATION, JANUARY 2024

ORGANIC CHEMISTRY–I

Time: 03 HOURS **Maximum Marks: 50**

SECTION – A

1. Answer any FOUR of the following questions. **4×5=20**

a. What are diasteriomers? Mention the types of diasteriomers with examples.

b. Describe conformational analysis in 1, 2-disubstituted cyclohexane.

c. What are the factors affecting the stability of carboanions? Explain.

d. Write a note on nucleophilic addition reactions.

e. What is aldol condensation? Give an example with a mechanism.

SECTION – B

2. Answer any TWO of the following questions. **2×9=18**

a. Discuss geometrical isomerism due to -C=C- bond and its classifications.

b. **i.** Why tertiary carbocation is more stable than secondary and primary carbocation? Explain with examples.

ii. Give any three methods for the formation of free radicals.

c. What is the Chugaev elimination reaction? Explain its mechanism with a suitable example.

SECTION – C

3. Answer the following question. **12**

a. **i.** Explain the diasterioselective reactions with suitable examples.

ii. Discuss SRN^1 mechanism in detail.

iii. Explain the Hell-Volhard-Zelinsky reaction with the help of a mechanism.